

# Storage Devices -- Guided Notes

Unit 2.3 | CompTIA Network+ N10-009 Obj. Core 1 3.4 | N10-008

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

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*Complete each question using the lesson reading. Write in the blank or answer in the space provided.*

## Question 1

A 5400 RPM HDD is typically used in \_\_\_\_\_ drives due to lower heat and power consumption. A 7200 RPM HDD provides \_\_\_\_\_ performance and is common in \_\_\_\_\_ storage applications. S.M.A.R.T. stands for \_\_\_\_\_ and is a built-in system that monitors \_\_\_\_\_ health and predicts potential failure.

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## Question 2

A \_\_\_\_\_ sound coming from a hard drive (HDD) is a symptom of mechanical failure -- a read/write head is likely contacting the \_\_\_\_\_. The correct immediate response is to \_\_\_\_\_ all important data before the drive fails completely.

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## Question 3

SATA III has a maximum transfer speed of approximately \_\_\_\_\_ MB/s. An NVMe SSD using PCIe 4.0 can reach speeds of up to \_\_\_\_\_ MB/s. This means NVMe is roughly \_\_\_\_\_ times faster than the fastest SATA SSD for sequential reads.

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## Question 4

An M.2 SSD with the designation 2280 is \_\_\_\_\_ mm wide and \_\_\_\_\_ mm long. The M.2 form factor connects directly to the motherboard using an \_\_\_\_\_ slot, eliminating the need for a \_\_\_\_\_ cable. M.2 drives can use either the SATA or \_\_\_\_\_ protocol depending on the type.

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## Question 5

RAID 0 stripes data across \_\_\_\_\_ or more drives for improved \_\_\_\_\_. It provides \_\_\_\_\_ (does/does not) redundancy. RAID 1 mirrors data across \_\_\_\_\_ drives, providing \_\_\_\_\_ redundancy. If one drive fails in RAID 1, the data is \_\_\_\_\_ from the mirror.

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### Question 6

RAID 5 requires a minimum of \_\_\_\_\_ drives and distributes \_\_\_\_\_ data across all drives. If one drive fails, data can be reconstructed using the \_\_\_\_\_ information. RAID 5 provides both performance and redundancy but \_\_\_\_\_ (can/cannot) survive a two-drive failure.

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### Question 7

FAT32 has a maximum individual file size of \_\_\_\_\_ GB. This means you \_\_\_\_\_ (can/cannot) store a 30 GB video file on a FAT32-formatted drive. NTFS and exFAT do not have this limitation. GPT partition style is required when the drive is larger than \_\_\_\_\_ TB and supports up to \_\_\_\_\_ primary partitions.

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### Question 8 -- Real World Application

REAL WORLD: A small business owner asks you to set up storage for a file server that needs to survive a single drive failure, maximize storage efficiency, and hold files larger than 4 GB. They have four identical 4 TB drives available. Recommend a RAID level, explain your reasoning, calculate the usable storage, and specify the required file system.

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*Download the answer key from the class server:*

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wget http://[SERVER]/resources/network-engineer/2-3-storage-devices/2-3-storage-devices-answer-key.pdf
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